

Practical No: 2

A) **Aim:** Write a program to implement predefined packages (java.util, java.io, etc.).

1. Using java.util.Scanner for input:-

```
import java.util.Scanner;
public class input {
    Run | Debug
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.print(s: "Enter your Name: ");
        String name = sc.nextLine();
        System.out.println("Welcome " + name + " How are You ?");
    }
}
```

```
Enter your Name: Adnan
Welcome Adnan How are You ?
PS D:\Practicals SYCS\SYCS\j
```

2. Using java.lang.Math for Calculation:-

```
import java.util.Scanner;
import java.lang.Math;

public class M{
    Run | Debug
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        System.out.print(s: "Enter a number: ");
        double a = sc.nextDouble();
        System.out.println("Square root of a: " + a + " is " + Math.sqrt(a));
        double c = sc.nextDouble();
        double b = sc.nextDouble();
        System.out.println("Power of c^b: " + c + " , " + b + " is " + Math.pow(c, b));
        System.out.println("PI Value of " + Math.PI);
    }
}
```

```
Enter a number: 70
Square root of a: 70.0 is 8.366600265340756
87 2
Power of c^b: 87.0 , 2.0 is 7569.0
PI Value of 3.141592653589793
PS D:\Practicals SYCS\SYCS\javaPractical>
```

3. Using java.time.LocalDate for time:-

```
import java.time.LocalDate;

public class DateExample{
    Run | Debug
    public static void main(String[] args){
        LocalDate today = LocalDate.now();
        System.out.println("Today's date is: " + today);
    }
}
```

ical_9b17b1aa\bin' 'DateExamp
Today's date is: 2026-07-19
PS D:\Practicals SYCS\SYCS\ja

4. Using java.io.File:-

```
import java.io.File;
public class file{
    Run | Debug
    public static void main(String[] args) {
        File file = new File(pathname: "text.txt");
        if (file.exists()) {
            System.out.println("File Exist: " + file.getName());
        } else {
            System.out.println(x: "File Not Found");
        }
    }
}
```

ical_9b17b1aa\bin
File Not Found
PS D:\Practicals

5. Using java.net.URL for [URL](#):-

```
import java.net.URL;

public class NetEx {
    Run | Debug
    public static void main(String[] args) throws Exception {
        URL url = new URL(spec: "https://google.com");

        System.out.println("Host: " + url.getHost());
        System.out.println("Protocol: " + url.getProtocol());
    }
}
```

ical_9b17b1aa\bin
Host: google.com
Protocol: https
PS D:\Practicals S

6. Using java.security;-

```
import java.security.MessageDigest;

public class message {
    Run | Debug
    public static void main(String[] args) throws Exception {
        String password = "add@123";
        MessageDigest md = MessageDigest.getInstance("SHA-256");
        byte[] hash = md.digest(password.getBytes());

        StringBuilder hex = new StringBuilder();
        for (byte b: hash) {
            hex.append(String.format("%02x",b));
        }
        System.out.println("SHA-256 Hash: " + hex.toString());
    }
}

SHA-256 Hash: 1d983eff6990495c0534a0e4e31c809152eb7c7369c96bd1dd622b48d3b6f15a
```

B) Write a Program to create and use a user defined packages.

1. Create a package src.myapp.utils.Mathutils.java:

```
package myapp.utils;

public class Mathutils {
    public static int square(int X){
        return X * X;
    }

    public static int cube(int X){
        return X * X *X;
    }
}
```

2. Create another file to import packages:

```
import myapp.utils.Mathutils;

public class AppMain {
    Run | Debug
    public static void main(String[] args) {
        System.out.println("Square of 4: " + Mathutils.square(X: 4));
        System.out.println("Cube of 4: " + Mathutils.cube(X: 4));
    }
}

Square of 4: 16
Cube of 4: 64
```

C) **Aim:** Write a program to demonstrate the use of access specifiers (Public, Private, Protected).

1. Public Specifier: -

```
public class online {  
    public String name = "Jerry";  
  
    public void show(){  
        System.out.println("Hello, " + name);  
    }  
    Run | Debug  
    public static void main(String[] args) {  
        online obj = new online();  
        obj.show();  
    }  
}
```

```
PS D:\Practicals> java -cp .\src\online\online.jar  
Hello, Jerry
```

2. Private Specifier: -

```
public class online {  
    private int bal = 100;  
    private void show(){  
        System.out.println("Balance: " + bal);  
    }  
  
    Run | Debug  
    public static void main(String[] args) {  
        online obj = new online();  
        obj.show();  
    }  
}
```

```
PS D:\Practicals> java -cp .\src\online\online.jar  
Balance: 100
```

3. Protected Specifier: -

```
public class online {  
    protected void sound(){  
        System.out.println("Animal Sound");  
    }  
    Run | Debug  
    public static void main(String[] args) {  
        online obj = new online();  
        obj.sound();  
    }  
}
```

```
PS D:\Practicals> java -cp .\src\online\online.jar  
Animal Sound
```